

Methanol at Water–TiO₂ Interfaces: Free Energies of Water and Methanol Dissociation

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Cite This: *ACS Catal.* 2026, 16, 1468–1478

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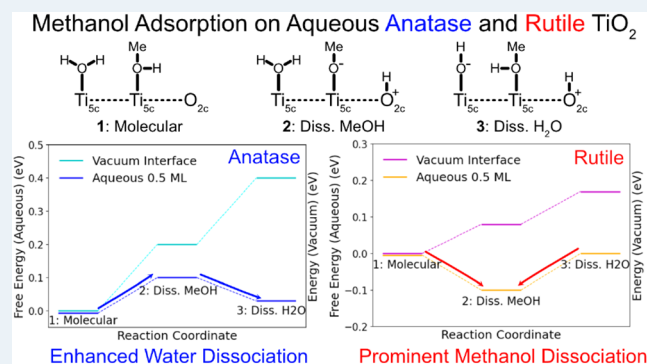
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ABSTRACT: Methanol adsorption on TiO₂ surfaces has long been studied due to its role in enhancing photocatalytic hydrogen evolution, yet how it modulates surface chemistry under aqueous conditions remains little understood. Using molecular dynamics with an *ab initio*-based deep neural network potential, we find that methanol adsorption induces markedly different effects on the aqueous surfaces of anatase and rutile, the two common phases of TiO₂. In anatase, methanol adsorption significantly enhances water dissociation, which is otherwise rare at the neat water interface. This enhancement arises from an alternative dissociation pathway mediated by surface-bound methoxyl groups. In contrast, methanol adsorption tends to suppress water dissociation on rutile, replacing it with thermodynamically favored methanol dissociation. Overall, methanol adsorption in an aqueous environment alters not only the availability of key reactive intermediates involved in hydrogen evolution but also the hydrogen source, which turns out to be primarily methanol on rutile, whereas both water and methanol are consumed on anatase. These results provide mechanistic insights into the coupled roles of organic adsorbates and water at photocatalytic interfaces, with implications on how methanol enhances the activity of H₂ evolution.

KEYWORDS: TiO₂–Water Interface, Methanol Adsorption, Machine Learning Interatomic Potentials, Hole Scavengers, Photocatalytic Water Splitting



INTRODUCTION

Photocatalytic water splitting continues to attract a great deal of attention as a promising route toward sustainable energy production.^{1–6} Among various photoactive semiconductors, titanium dioxide (TiO₂) is considered a prototypical photocatalytic material due to its abundance, low toxicity, and environmental compatibility.^{7–11} Despite all these benefits, TiO₂ suffers from an inherent low efficiency in the overall photocatalytic water splitting process. To address this limitation, hole scavengers, also known as sacrificial agents, are commonly employed to enhance hydrogen production.^{12–15} In particular, methanol serves as an ideal candidate for such applications, as it can potentially be produced from renewable or waste-derived carbon sources, e.g., from biomass or toxic residues, and at the same time provides an easy oxidation route to CO₂ due to the absence of C–C bonds.^{16–19} Consequently, considerable effort has been devoted to understanding how methanol promotes hydrogen evolution, both experimentally and theoretically.^{20–28}

A key step in the promoting effect of methanol is its dissociation, as molecular methanol shows negligible photocatalytic activity prior to dissociation.^{29–31} Whether methanol spontaneously dissociates on TiO₂ has been a topic of extensive debate, especially for the most common rutile

(110) and anatase (101) surfaces.^{31–35} Questions also remain regarding the precise mechanism by which methanol enhances photocatalytic hydrogen production. It is widely agreed that water oxidation, a known bottleneck in water splitting,³⁶ is replaced by methanol oxidation, which scavenges photo-generated holes more effectively and thus inhibits electron–hole recombination.^{12–14,37–41} The remaining electrons are then used to drive hydrogen production by reducing the protons of bridging hydroxyls.^{42,43} Although it is known that hydrogen can be generated through the consumption of either methanol or water, depending on the concentration of methanol and the TiO₂ phase, understanding the underlying factors that govern this preference remains a challenge.^{14,44,45}

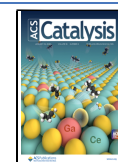
In this work, we seek to shed light on the nature of methanol adsorption on TiO₂ and its impact on the chemistry of aqueous TiO₂ interfaces using *ab initio*-based simulations.^{46,47} Previous computational studies of the adsorption and photo-

Received: October 21, 2025

Revised: December 6, 2025

Accepted: December 18, 2025

Published: December 24, 2025



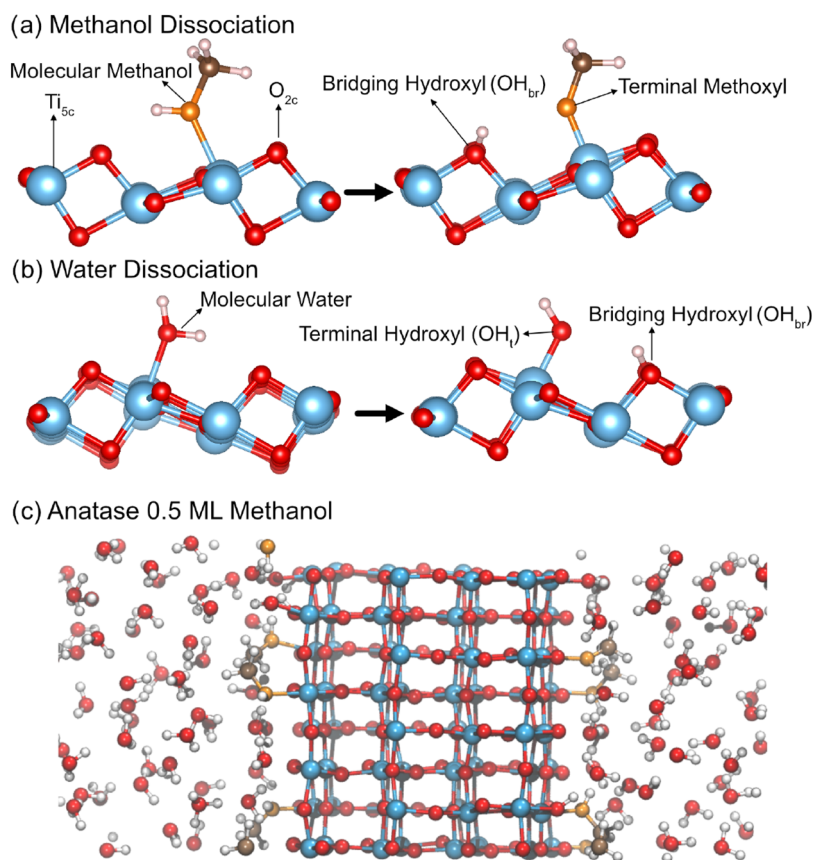


Figure 1. Ball and stick models of typical molecular and dissociated adsorption configurations of methanol (a) and water (b) at the aqueous anatase (101) interface. Undercoordinated surface Ti_{5c} and O_{2c} sites are marked with arrows. Water and methanol adsorb on the Ti_{5c} sites, with protons transferred to the O_{2c} sites upon dissociation. Key species include terminal methoxyls, terminal hydroxyls (OH_t), and bridging hydroxyls (OH_{br} or $O_{2c}H$) arising from either water or methanol dissociation. (c) Representative snapshot from DPMD simulations of the aqueous anatase (101) surface covered with 0.5 ML methanol.

chemistry of methanol on TiO_2 were performed under vacuum conditions, which do not fully capture the aqueous environments relevant to photocatalysis.^{25,26,31,33} To remedy this, we investigated the interfacial dynamics under aqueous conditions with preadsorbed methanol molecules on the TiO_2 surface. Although *ab initio* molecular dynamics (AIMD) is in principle well-suited for modeling such systems, its applicability is restricted by high computational cost. Here we construct instead an *ab initio*-based deep neural network interatomic potential (DP),⁴⁸ which effectively extends the accessible time scale of molecular dynamics simulations by orders of magnitude without sacrificing accuracy.^{49–51} This allows direct observation of rare but critical events, such as water and methanol dissociation, which often occur on much longer time scales than those accessible by AIMD.⁵²

Our DP-based molecular dynamics (DPMD) simulations reveal important differences in the adsorption modes and reaction (dissociation) pathways of methanol at the aqueous rutile and anatase interfaces. While rutile favors dissociative adsorption of methanol along with suppressed water dissociation, anatase favors molecular methanol adsorption accompanied by enhanced water dissociation. Such contrasting behaviors are governed by distinct hydrogen-bonding environments of the adsorbates, arising from intrinsic structural differences between the anatase and rutile phases. Ultimately, this modulates the free-energy landscape of complex reaction pathways and shifts the populations of key photoactive

intermediate species. These results provide mechanistic insights into the evolution of hydrogen on the surfaces of TiO_2 under complex aqueous conditions involving water and organic adsorbates.

RESULTS AND DISCUSSION

Equilibrium DPMD Simulations. The forces and energies used to train the DP were generated using the SCAN functional, which is known to be accurate for both water and TiO_2 .^{53–55} Details of our computational approach, including training and validation of the DPs and DFT calculations, are presented in the [Methods](#) section (see below) and the [Supporting Information \(SI\)](#). The interfacial TiO_2 systems considered in our study comprise anatase (101) and rutile (110) surfaces terminated with low (0.125 ML) or high (0.5 ML) coverages of adsorbed methanol, as well as bare surfaces, all interfaced with a 30 Å water slab. Both anatase (101) and rutile (110) surfaces are terminated with five-coordinated Ti atoms (Ti_{5c}) and two-coordinated O atoms (O_{2c}), as illustrated in [Figure 1](#). The water and methanol molecules adsorb at the Ti_{5c} sites and can dissociate by transferring a proton to a neighboring O_{2c} site, resulting in terminal methoxyl or hydroxyl groups (OH_t) together with bridging hydroxyls ($O_{2c}H$ or OH_{br}). [Figure 1c](#) shows a representative DPMD snapshot of the aqueous anatase (101) surface modified with 0.5 ML of methanol.

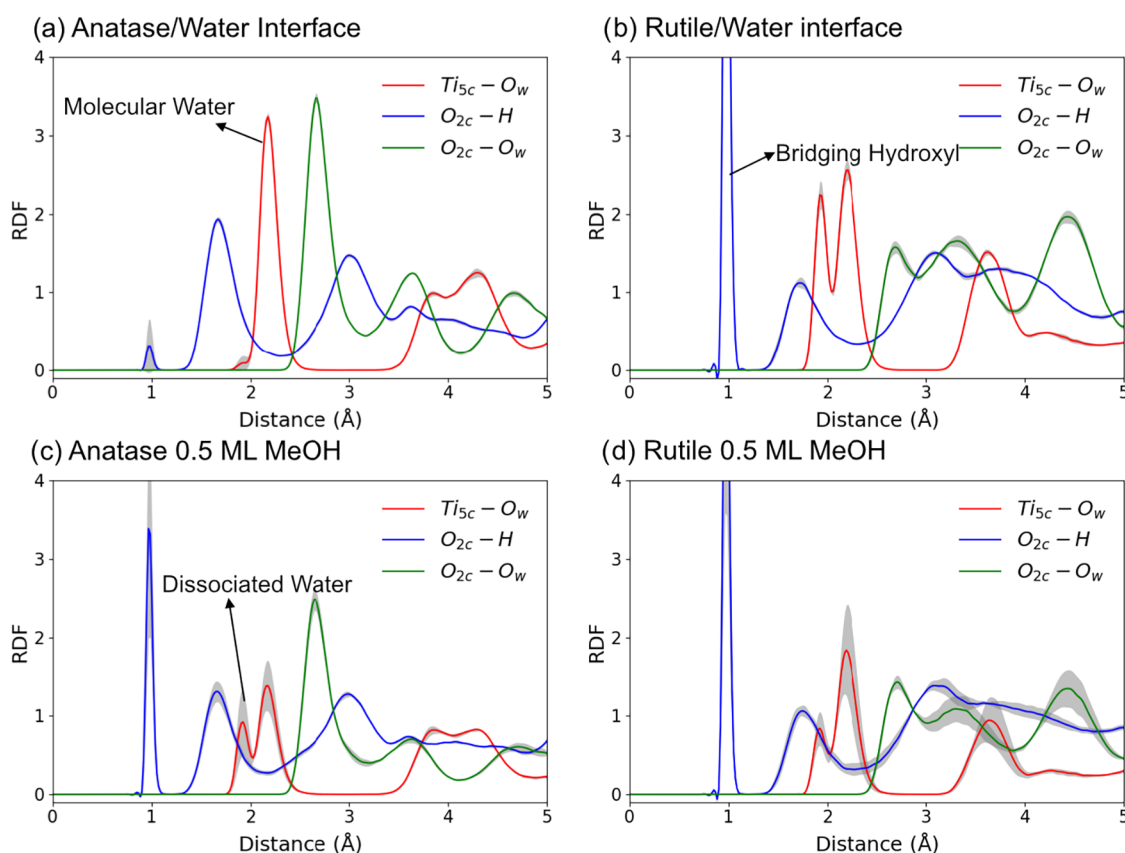


Figure 2. Radial distribution functions (RDFs) of $Ti_{5c}-O_w$, $O_{2c}-H$, and $O_{2c}-O_w$ pairs at the bare anatase–water (a) and rutile–water (b) interfaces and at the methanol (0.5 ML) covered anatase–water (c) and rutile–water (d) interfaces. The shaded regions indicate standard deviations across the three trained DPs. The peak at 1.9 Å in the red curves corresponds to dissociated water (OH_t), while the peak at 2.1 Å represents adsorbed molecular water. In the blue curves, the peak at 1 Å corresponds to hydroxylated O_{2c} sites (OH_{br}), formed via either water or methanol dissociation. From the RDF profiles, anatase shows an increase in water dissociation at high coverage, whereas rutile exhibits a noticeable decrease while largely retaining the OH_{br} peak.

The equilibrium structures of the interfacial systems were determined by averaging the results of three independent DPMD trajectories of 10 ns duration each. Figure 2a–d show the radial distribution functions (RDFs) of Ti_{5c} –water oxygen (O_w), O_{2c} –H, and O_{2c} – O_w pairs for the aqueous interfaces of bare and methanol-covered (0.5 ML) anatase (101) and rutile (110), with shaded regions indicating the standard deviation among the three DPs. As observed in previous studies, the bare anatase–water interface seldom exhibits water dissociation, and most water molecules remain in the molecular form (peak at ~ 2.1 Å in the red curve in Figure 2a).⁴⁹ In contrast, the rutile–water interface is significantly more reactive, showing a pronounced dissociated water (OH_t) peak at 1.9 Å and a hydroxylated O_{2c} (OH_{br}) peak at 1.0 Å.⁵⁰ This behavior is quantified by the fraction of water dissociation, calculated as the number of OH_t relative to the number of available surface Ti_{5c} sites for water adsorption, averaged over three equilibrium DPMD trajectories. As shown in Table 1, rutile clearly exhibits a much higher water dissociation fraction (31.6%) compared to anatase (1.4%), both values showing reasonable agreement with previous work.

Having established the baseline behavior of water dissociation at bare TiO_2 –water interfaces, we next focus on the structural changes induced by adsorbed methanol. For anatase, the RDFs show a pronounced increase in both the OH_{br} and the OH_t peaks (Figure 2c), which are barely visible in the absence of methanol. This indicates a significant enhancement

Table 1. Average Fractions of Methanol and Water Dissociation and O_{2c} Hydroxylation Calculated from Three Independent Equilibrium DPMD Trajectories^a

surface-methanol coverage	methanol dissociation	O_{2c} hydroxylation	water dissociation
anatase-0.0 ML	N/A	$1.4 \pm 1.4\%$	$1.4 \pm 1.4\%$
anatase-0.125 ML	$1.2 \pm 1.7\%$	$0.8 \pm 0.1\%$	$0.7 \pm 0.3\%$
anatase-0.5 ML	$3.0 \pm 1.5\%$	$15.5 \pm 6.0\%$	$26.6 \pm 13.2\%$
rutile-0.0 ML	N/A	$31.6 \pm 1.9\%$	$31.6 \pm 1.9\%$
rutile-0.125 ML	$10.4 \pm 11.3\%$	$31.0 \pm 6.2\%$	$31.8 \pm 6.3\%$
rutile-0.5 ML	$40.0 \pm 15.5\%$	$24.3 \pm 11.1\%$	$16.4 \pm 5.6\%$

^aFractions are normalized by the number of corresponding available sites. For example, at 0.5 ML coverage, there are 16 O_{2c} sites along with 8 Ti_{5c} adsorption sites for both water and methanol. In this case, the O_{2c} hydroxylation fraction should equal the average of methanol and water dissociation fractions, provided no methanol desorption occurs.

in water dissociation. As shown in Table 1, high methanol coverage induces an increase in water dissociation and O_{2c} hydroxylation fractions, while the methanol dissociation fraction remains relatively small. These results suggest that water likely serves as one of the main sources of OH_{br} and that the adsorption of methanol promotes the dissociation of water, nearly approaching the level seen in rutile. This enhancement is expected to facilitate hydrogen evolution, as OH_{br} serves as a key intermediate in the process.

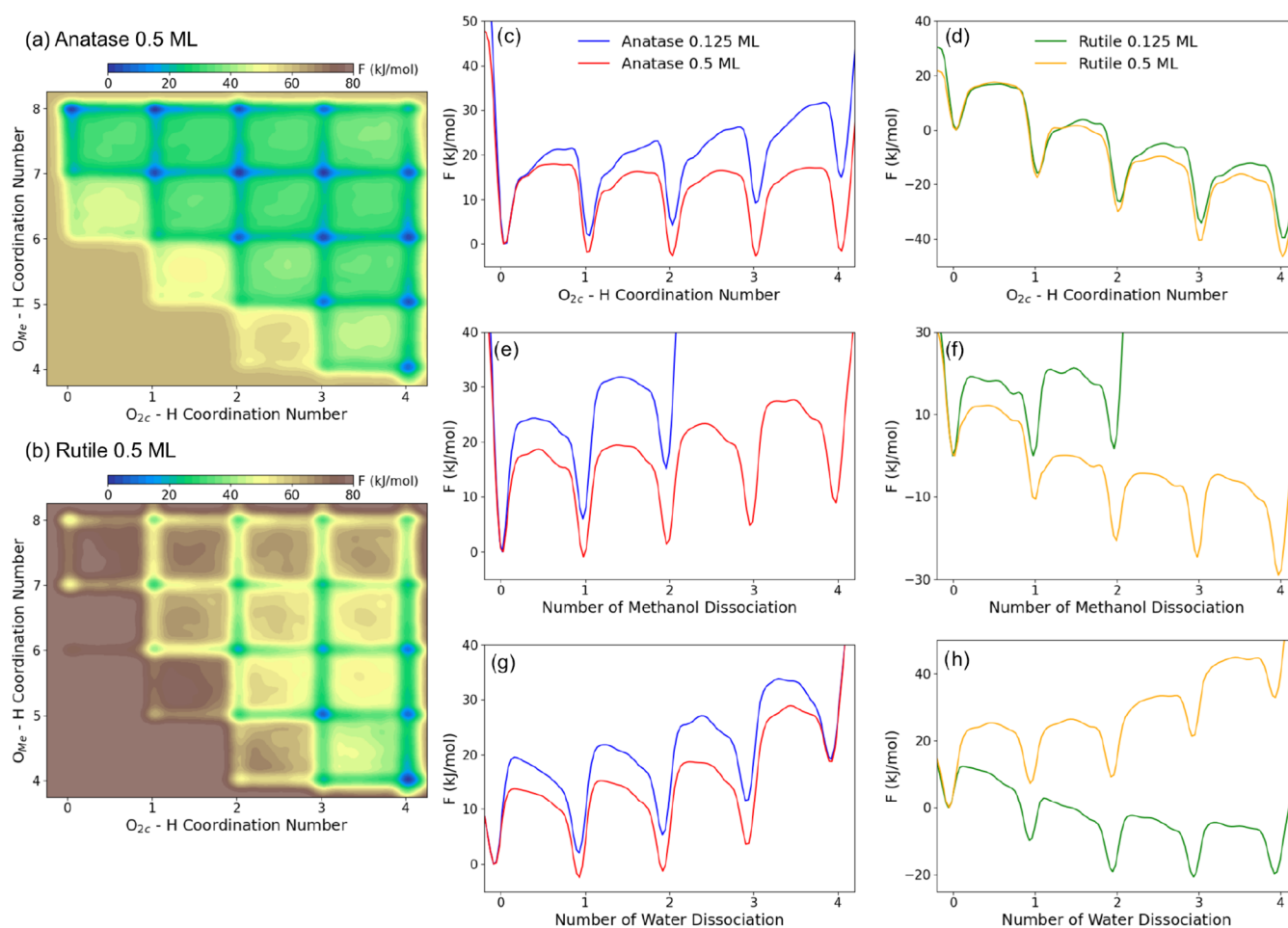


Figure 3. Two-dimensional projections of the free energy surfaces (FES) from the metadynamics simulations of the high-coverage anatase (a) and rutile (b) systems, projected onto the methoxyl oxygen (O_{Me})–H and O_{2c} –H coordination number CVs. The colors represent free energy values, with darker blue indicating lower energy. Anatase primarily populates off-diagonal states, suggesting prominent water dissociation. Rutile, on the other hand, predominantly populates diagonal states, where O_{2c} hydroxylation is driven purely by methanol dissociation. From the same metadynamics calculations, one-dimensional free energy projections for O_{2c} hydroxylation (c, d), methanol dissociation (e, f), and water dissociation (g, h) are presented. On anatase, 0.5 ML methanol coverage lowers the free energy of all three processes, suggesting an overall enhancement in its reaction dynamics. On rutile, methanol dissociation becomes more favorable, while water dissociation is suppressed. The O_{2c} hydroxylation profile, however, remains largely unaffected by methanol coverage.

The 0.5 ML methanol-covered rutile–water interface, on the other hand, shows quite different behavior (Figure 2d). While the OH_{br} peak looks largely unchanged, the relative intensity of the molecular and dissociated water peaks changes in a way that indicates diminished water dissociation. As shown in Table 1, the water dissociation fraction in fact decreases substantially, while the change in the O_{2c} hydroxylation fraction is rather modest. The dissociation fraction of methanol is substantially higher than that of water at 0.5 ML coverage, indicating a shift in the source of O_{2c} hydroxylation from water to methanol. This suggests that, contrary to anatase, methanol is likely to be the dominant source of hydrogen production on rutile. For the rutile surface, we also observed desorption of approximately 50% of the initially adsorbed methanol during our 10 ns simulations, leading to the formation of water-solvated species (Figures S7–S8 in SI). Consequently, the reported high-coverage rutile system actually represents a system with intermediate methanol coverage. To confirm these results, we also calculated the free energy profiles of methanol adsorption/desorption on/from anatase and rutile, which are presented in the SI.⁵⁶ As expected, the free energy of methanol

desorption is significantly smaller for rutile, accompanied by lower energy barriers, which explains the prominent desorption events observed in the DPMD simulations.

Free Energies of Water and Methanol Dissociation.

To substantiate the trends shown in Table 1, we further investigated the free energy surface of water and methanol dissociation at the anatase and rutile interfaces using metadynamics with three collective variables (CVs): O_{2c} –H, methoxyl oxygen (O_{Me})–H, and O_w –H coordination numbers (see SI for details). These CVs represent the populations of OH_{br} , dissociated methanol, and dissociated water, allowing us to track all dissociation reactions throughout the metadynamics. Two-dimensional projections of the free energy surfaces along the O_{2c} –H and O_{Me} –H coordination number axes are shown in Figure 3a,b for the 0.5 ML methanol coverage systems. The blue regions in the plot represent energetically favorable coordination states, with darker hues indicating lower free energy. Although the O_w –H coordination number is not explicitly shown, its contribution can be inferred by tracking the difference between hydroxylated O_{2c} sites and dissociated methanol counts. Briefly, points along the diagonal

correspond to configurations in which the OH_{br} groups originate entirely from the dissociation of methanol, while off-diagonal regions reflect a greater contribution from water dissociation. Both anatase and rutile exhibit prominent O_{2c} hydroxylation at high methanol coverage, but with distinct trends. Rutile predominantly populates a diagonal state with four dissociated methanol, suggesting that O_{2c} hydroxylation is primarily driven by methanol dissociation. In contrast, anatase favors off-diagonal regions, indicating a greater contribution from water dissociation.

It is also instructive to project the free energy surfaces onto one-dimensional profiles for each CV and compare the results for low and high methanol coverages. Figure 3c,d show the free energy as a function of sequential O_{2c} hydroxylation, in which the coordination number corresponds to the number of formed OH_{br} species. At low methanol coverage, the system should reproduce the behavior seen at the bare water interface, as 0.125 ML does not induce meaningful changes in dissociation fractions. Under this condition, we see that O_{2c} hydroxylation is disfavored in anatase but strongly favored in rutile, consistent with the results in Table 1. At 0.5 ML coverage, the free energy curve for anatase shifts downward, indicating enhanced O_{2c} hydroxylation. In contrast, rutile shows minimal change between low and high methanol coverage, supporting the observation that O_{2c} hydroxylation in rutile is largely independent of methanol coverage. Figure 3e,3f show the free energy as a function of the dissociated methanol count. In both anatase and rutile, methanol dissociation is more favorable at high coverage. This effect is significantly more pronounced in rutile, as evidenced by the substantial decrease in its free energy profile. In Figure 3g,3h, a very different trend emerges for the dissociation of water: while high methanol coverage enhances the dissociation of water in anatase, this is strongly suppressed in rutile.

From these free energy profiles, we can obtain an additional independent estimate of the fractions introduced in Table 1. This is given by

$$\text{average fraction} = \frac{\sum_i f_i \exp(-\beta A_i)}{\sum_j \exp(-\beta A_j)} \quad (1)$$

where the summation is taken over all states *i*, ranging from 0 to 4, which represent the observed numbers of O_{2c}H, methanol dissociations and water dissociations, with *f_i* and *A_i* representing the fraction and free energy of state *i*. Due to the upper bounds imposed on the coordination number CVs, the resulting fractions may be underestimated. However, as shown in Table 2, these values generally agree well with those derived from the equilibrium simulations and provide a more robust estimate that is less susceptible to sampling limitations. Our metadynamics simulations confirm the trend that 0.5 ML methanol coverage enhances methanol dissociation, O_{2c}

hydroxylation, and water dissociation in anatase. For rutile, they confirm that water dissociation is strongly suppressed at 0.5 ML, indicating a change from water to methanol as the dominant source of O_{2c} hydroxylation. It should be noted that this effect is even more pronounced in metadynamics simulations, where methanol desorption is prohibited, resulting in a water dissociation fraction of less than 2%. Because these results provide a clear trend in interfacial dissociation dynamics for both anatase and rutile, and show good agreement with the DPMD simulations, they serve as the basis for all discussions throughout this paper.

Understanding the Difference between Anatase and Rutile. To gain insight into the different behaviors of anatase and rutile in response to methanol adsorption, we performed equilibrium DPMD simulations starting from configurations in which all methanol molecules were initially dissociated while all water remained in the molecular state. We then monitored the populations of OH_v, OH_{br}, and molecular methanol along the trajectory, as shown in Figure 4a,b. In anatase, all terminal methoxyls were converted to molecular methanol within 1 ns, accompanied by a marked increase in the number of OH_v, indicating a rapid dissociation of adsorbed water. The conversion of methoxyl groups to methanol occurred by proton transfer either from OH_{br} sites or from adjacent water molecules. Based on the dynamics in Figure 4a, both pathways appear equally favorable, suggesting that methanol adsorption creates a new channel for water dissociation, ultimately increasing its fraction. This behavior was also confirmed by AIMD simulations, which demonstrated the same water dissociation channel mediated by methoxyl groups on a similar time scale (see details in the SI).

A quite different behavior is observed for rutile, which exhibits a conversion of ~50% methoxyl groups to methanol on a time scale of approximately 2 ns (Figure 4b). This transformation occurs exclusively through recombination with OH_{br}, while the alternative water dissociation pathway appears to be strongly disfavored. Considering that methanol desorption was not observed in the particular DPMD trajectory shown in Figure 4b, the absence of OH_t species suggests that a true 0.5 ML methanol coverage environment suppresses water dissociation. This also implies that the relatively high water dissociation fraction reported in Table 1 is likely a consequence of the desorption of methanol. In other words, OH_{br} formation at high coverage of methanol is driven mainly by dissociation of methanol rather than dissociation of water.

The above results indicate that in anatase, methanol adsorption introduces a new water dissociation pathway mediated by methoxyl groups, as illustrated in Figure 5a. Here, 1 is the initial state, where both water and methanol are adsorbed in molecular form at adjacent Ti_{5c} sites, 2 is the intermediate state where methanol is dissociated to form a terminal methoxyl and OH_{br}, and 3 is the final state where the methoxyl has captured a proton from the adjacent water molecule resulting in OH_v, OH_{br}, and molecular methanol. While water dissociation in the absence of methanol adsorption proceeds directly from 1 to 3, our simulations suggest that the same process may instead be mediated by 2, i.e., 1 → 2 → 3. Similar mechanisms have been proposed in past experimental studies, commonly referred to as the indirect oxidation pathway, in which redox reactions occur across adsorbates rather than directly from TiO₂.^{14,44} To quantify the free energies associated with methanol dissociation (1 → 2)

Table 2. Average Fractions of Methanol and Water Dissociation and O_{2c} Hydroxylation Calculated from Metadynamics Free Energy

surface-methanol coverage	methanol dissociation	O _{2c} hydroxylation	water dissociation
anatase-0.125 ML	4.3%	3.4%	3.2%
anatase-0.5 ML	12.3%	14.1%	15.0%
rutile-0.125 ML	39.7%	24.3%	21.8%
rutile-0.5 ML	47.5%	24.5%	1.2%

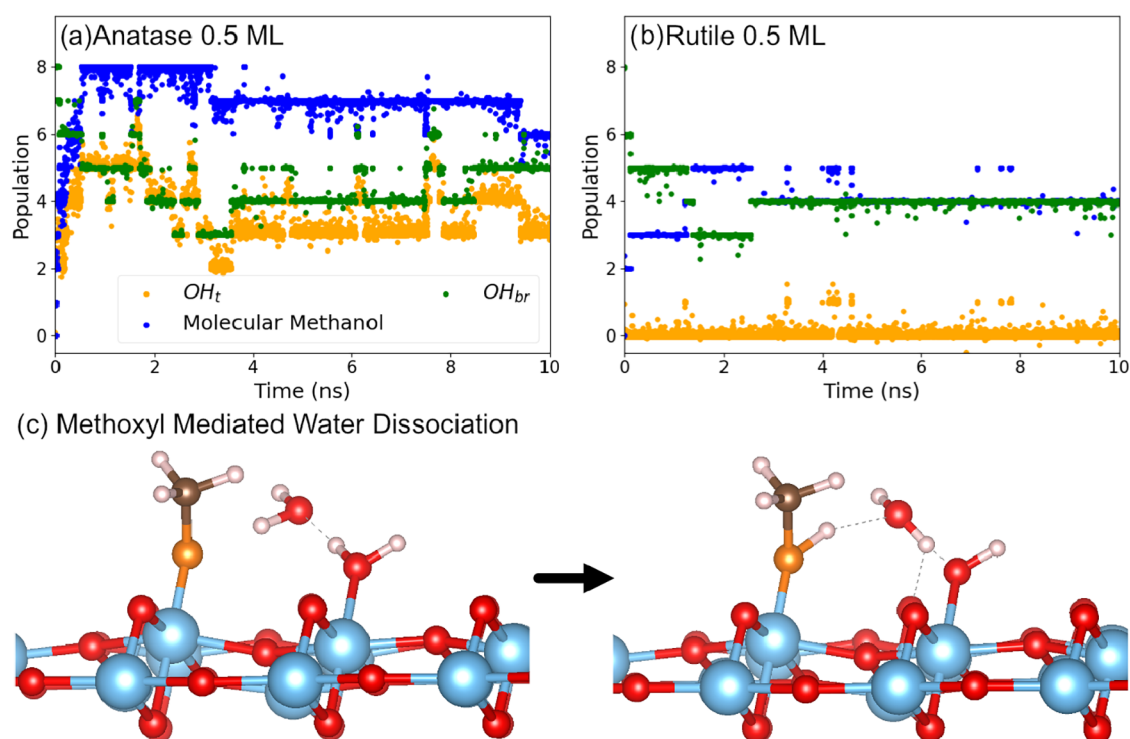


Figure 4. Time evolution of the populations of OH_t (orange), OH_{br} (green), and molecular methanol (blue) on anatase (a) and rutile (b) surfaces from DPMD simulations initiated with a configuration including eight dissociated methanol adsorbed on the surface; blue dots start from zero, while green dots start from eight. Populations are computed using the smooth and continuous coordination number formalism introduced in the SI. Noninteger populations indicate ongoing dissociation or recombination processes. On anatase, rapid methoxyl protonation is observed, accompanied by either water dissociation or O_{2c} dehydroxylation. Rutile shows negligible water dissociation and a coexistence of both dissociated and molecular methanol. The representative snapshots in panel (c) illustrate water dissociation through proton transfer to an adjacent methoxyl, mediated by a second layer water molecule.

and methoxyl-mediated water dissociation ($2 \rightarrow 3$) under aqueous conditions, we performed umbrella sampling and metadynamics simulations with methodological details provided in the SI.

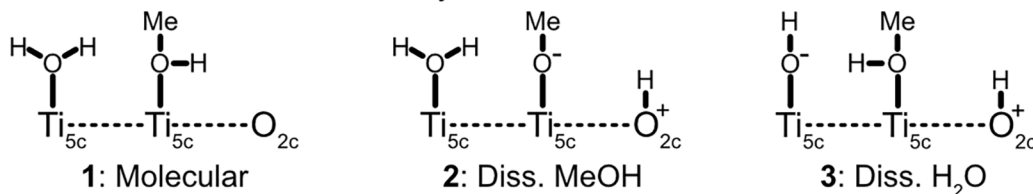
The free energy profiles of the two individual reactions are shown in Figure 5b,c, in which the reactions proceed from negative to positive reaction coordinate values. We note that these profiles represent a single dissociation reaction and would likely shift in the presence of multiple coadsorbed dissociated species as indicated in Figure 3. For methanol dissociation ($1 \rightarrow 2$), it appears that the process is thermodynamically disfavored in anatase, with a positive change in free energy of approximately 7 kJ mol^{-1} , while dissociated methanol is more stable by approximately 10 kJ mol^{-1} in rutile. On the other hand, methoxyl-mediated water dissociation ($2 \rightarrow 3$) shows the opposite trend: the reaction is favored in anatase but not in rutile. These results are summarized in Figure 5d,e, which show the relative free energies of states 1, 2 and 3 at the aqueous anatase and rutile interfaces, and for comparison, under vacuum conditions with a single water and methanol coadsorbed on one surface. Clearly, both water and methanol dissociation processes are energetically uphill in vacuum since the formation of charge-separated species is unfavorable in the absence of solvent stabilization. Although all dissociated states are stabilized in an aqueous environment, state 3 exhibits much greater solvent stabilization in anatase, whereas the effect is rather modest in rutile.

To clarify the origin of this difference, we analyzed the hydrogen-bonding (H-bonding) environment of the charged

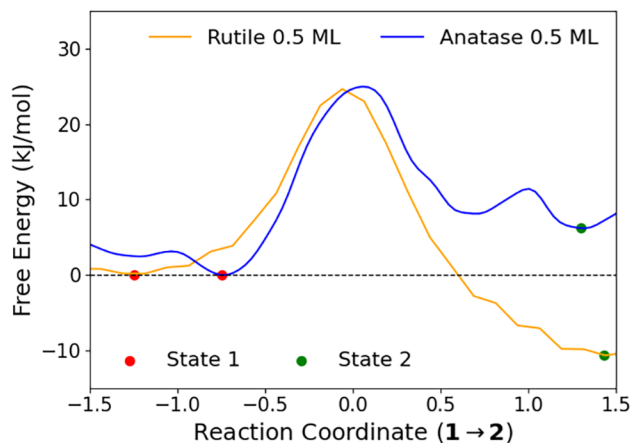
adsorbate in state 3, namely OH_t . An H-bond is defined as having donor (D)-acceptor (A) distances shorter than $3.5 \text{ \AA}$ and D-H-A angles larger than 120° . To sample configurations, we performed 3 ns of equilibrium DPMD with the system restrained at the free energy minimum of state 3, and collected H-bonding statistics for OH_t species. Subsequently, these data were binned into histograms shown in Figure 5f,g. Here, shorter O-H distances and larger O-H...O angles indicate stronger H-bonds. From the histograms, it appears that anatase provides a significantly more favorable H-bonding environment for OH_t , consequently lowering the free energy of state 3. In fact, anatase exhibits longer Ti_{5c} - Ti_{5c} distances,⁵⁷ allowing solvent molecules to approach charged adsorbates such as OH_t more closely and form stronger H-bonds. In rutile where Ti_{5c} - Ti_{5c} distances are shorter,⁵⁷ solvent molecules experience greater steric hindrance from the CH_3 groups of adsorbed methanol, which limits their ability to form favorable H-bonds with adsorbates and simultaneously causes significant disruptions of the H-bond network near the surface. These solvation effects, coupled with the energetics of charge separation associated with dissociation, form the basis of the free energy profile shown in Figure 5.

Overall, the free energy profiles in Figures 5d,e provide valuable information on the reasons why the two phases of TiO_2 exhibit markedly different behaviors under methanol adsorption. In anatase, the methoxyl groups are thermodynamically unstable. As a result, once formed, they rapidly return to the molecular state by abstracting a proton from either OH_{br} or adsorbed water. This introduces an additional channel for water dissociation, which ultimately increases its

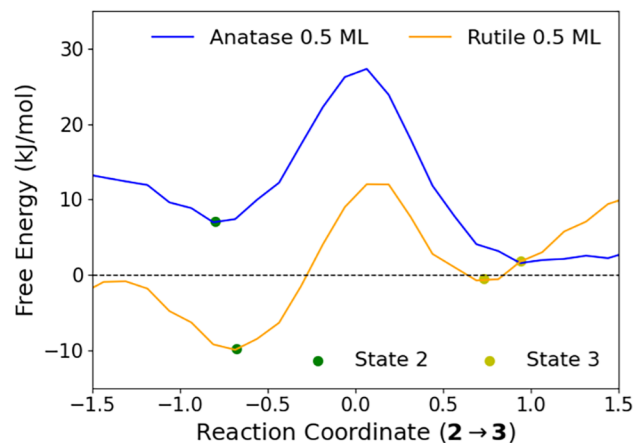
(a) Alternative Water Dissociation Pathway



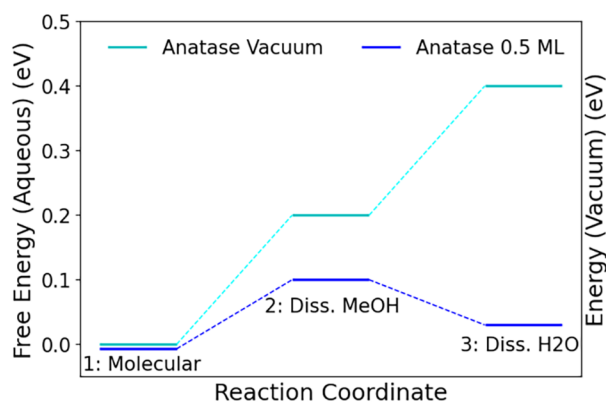
(b) Methanol Dissociation (1 → 2)



(c) Methoxyl Mediated Water Dissociation (2 → 3)



(d) Anatase Energy Profile



(e) Rutile Energy Profile

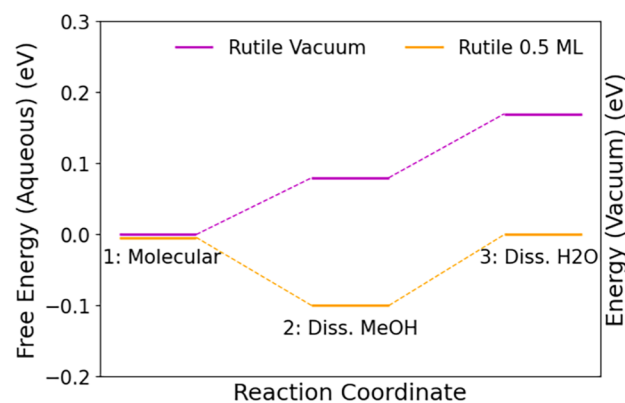
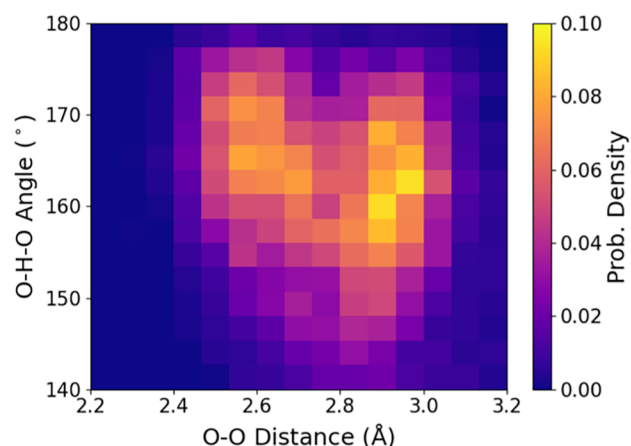
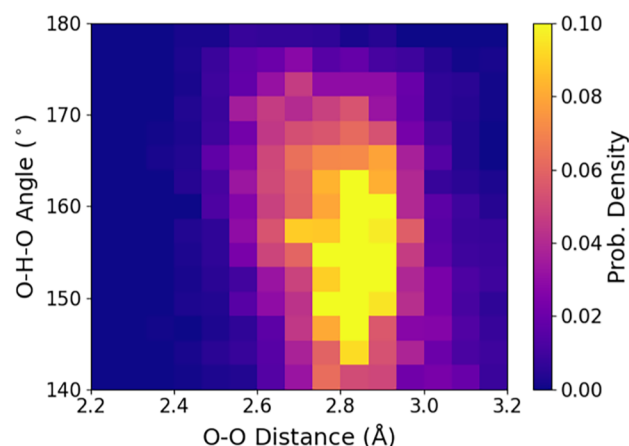
(f) Anatase 0.5 ML OH_i H-bond(g) Rutile 0.5 ML OH_i H-bond

Figure 5. (a) Initial (1), intermediate (2) and final (3) states of the alternative water dissociation pathway proposed in this work. The reaction consists of two steps: initial methanol dissociation (1 → 2) and methoxyl induced water dissociation (2 → 3). (b, c) Free energy profiles of 1 → 2 (left) and 2 → 3 (right) for the anatase (blue) and rutile (orange) interfaces along the corresponding reaction coordinates; while the first reaction step is favored in rutile, the second is favored in anatase. Red, green, and yellow dots represent states 1, 2, and 3, respectively, with the zero of free energy set at state 1. (d, e) Relative (free) energies of 1, 2, and 3 at the vacuum (aqueous) anatase (left) and rutile (right) interfaces. (f, g)

Figure 5. continued

Distribution of O–O distances and O–H...O angles characterizing the H-bonding environment experienced by the OH_t species at the aqueous anatase (left) and rutile (right) interfaces. Anatase provides a more favorable H-bonding environment for OH_t, characterized by shorter O–O distances and larger O–H...O angles.

occurrence. In contrast, methanol dissociation is strongly favored in rutile, even more so than water dissociation, thereby suppressing the latter. Even when water dissociation occurs in rutile (1 → 3), methanol can readily donate its proton to OH_t, generating methoxyl and molecular water (3 → 2).

We also note that the changes in the dissociation free energy landscape suggest changes in the effective pK_a values of surface species. In anatase, the more pronounced hydroxylation of O_{2c} sites in the presence of adsorbed methanol (0.5 ML) suggests an increase of the basicity of these sites, which is also consistent with the increase in water dissociation. In rutile, instead, O_{2c} hydroxylation appears to remain similar to that in pure water while water dissociation decreases and is essentially replaced by methanol dissociation. This suggests that methanol has a slightly lower pK_a than adsorbed water on rutile and the basicity of O_{2c} sites remains unchanged regardless of methanol adsorption. However, precise values of the pK_a would require further investigation.

CONCLUSIONS

This work demonstrates the intricate surface chemistry and reaction dynamics induced by adsorbed methanol at TiO₂–water interfaces. Specifically, our simulations reveal a clear trend in the mode of methanol adsorption at aqueous TiO₂ interfaces: anatase predominantly favors molecular adsorption, while rutile yields a mixture of molecular and dissociative adsorption, a trend consistent with recent experimental studies. Importantly, our results also show that surface-bound methoxyl groups on anatase introduce an alternative pathway for water dissociation, substantially increasing the populations of OH_{br} and OH_t, suggesting an improvement in photocatalytic hydrogen production. In contrast, rutile exhibits suppressed water dissociation and prominent dissociation of methanol at high coverage, suggesting a photocatalytic enhancement via methoxyl groups acting as more effective hole scavengers. Therefore, methanol is expected to be consumed primarily during the production of hydrogen in rutile, leading predominantly to methanol-derived oxidation products (formaldehyde and methyl formate), while anatase should generate fewer methanol-derived products and a larger amount of OH radicals, since both water and methanol are likely to be consumed during the evolution of H₂. Overall, these results indicate that the methanol induced enhancement in H₂ production occurs through different underlying mechanisms in anatase and rutile. Although further experimental verification is warranted, our predictions appear to agree with available experimental observations, lending support to the mechanistic insights presented in this work.

METHODS

Our deep learning interatomic potential (DP) was constructed using the DeepMD-kit package.⁵⁸ In this scheme, the total potential energy (E) of each configuration is expressed as the sum of the atomic energies E_i ($E = \sum_i E_i$), which are determined by feeding a descriptor constructed from local coordinate information within a cutoff radius r_c (here 6 Å) into

the deep neural network. A more detailed description of the DeepMD scheme is provided in ref 58. Given the large system of interest, the training set was generated on-the-fly using an iterative active learning scheme.⁵⁹ At each iteration, three independent DPs with different initializations were used to explore the configurational space, and configurations with high deviations in forces among the three DPs were selected to be included in the training set for the successive iteration. The active learning procedure was considered to have converged when the mean force deviation fell below 0.05 eV Å⁻¹ over a 100 ps-long DPMD run.

The labeling step was carried out with the Quantum Espresso package using the SCAN functional, which provides a good description of both water and TiO₂.^{53–55,60} The initial training set was adapted from two previous studies and included configurations of bulk anatase and rutile TiO₂, bulk liquid water, and anatase(101)–water and rutile(110)–water interfaces, totaling approximately 23,000 structures.^{49–51} To expand this set, we constructed additional systems comprising low (0.125 monolayer (ML)) and high (0.5 ML) coverage of adsorbed methanol on TiO₂–water interfaces, as well as solvated methanol systems. To efficiently sample rare events such as methanol dissociation and desorption in the training set, enhanced sampling was employed to generate relevant configurations. The final data set contained roughly 30,000 configurations, with further details on the active learning workflow provided in the Supporting Information (SI).

With the converged DPs, we performed DPMD simulations using the LAMMPS code interfaced with the DeepMD-kit.⁶¹ The anatase TiO₂(101) surface was modeled using a slab of 4 layers with a (1 × 4) surface unit cell, while for rutile TiO₂(110), we used a slab of 8 layers with a (2 × 4) surface unit cell. In both cases, each unit cell on both surfaces of the slab exposed eight 5-fold coordinated Ti atoms (Ti_{5c}) and eight two-coordinated O atoms (O_{2c}). Each TiO₂ slab was interfaced with a 30 Å thick slab of water at the experimental density. All simulations were carried out in the canonical ensemble at 300 K using the Nosé–Hoover chain thermostat.⁶² A time step of 0.5 fs was used, with H₂O replaced by D₂O to prevent integration errors without affecting statistical properties. Equilibrium DPMD was run with all three DPs for at least 10 ns. All reported equilibrium properties are averaged over trajectories simulated by three DPs. Enhanced sampling calculations were performed using the PLUMED package.⁶³ All enhanced sampling simulations were performed using only one of the three generated DPs.

To validate our models, we sampled configurations from 10 ns equilibrium DPMD trajectories and compared the resulting energy and force errors with those of the training set. For low coverage, the training set errors were $E_{\text{RMSE}} = 1.07$ meV atom⁻¹ and $F_{\text{RMSE}} = 207$ meV Å⁻¹; for high coverage, the values were $E_{\text{RMSE}} = 1.27$ meV atom⁻¹ and $F_{\text{RMSE}} = 205$ meV Å⁻¹ (Figure S2 in SI). These errors are comparable to those reported in previous studies of TiO₂ systems from our group.^{49–51} The validation set errors were even smaller, suggesting that our trained DPs are capable of generalizing to unseen yet relevant configurations. We further evaluated our

DPs by calculating the adsorption and dissociation energies of methanol at vacuum interfaces, which also showed good agreement with *ab initio* SCAN predictions. Lastly, we calculated the potential of mean force (PMF) for proton transfer from methanol to an adjacent O_{2c} site, finding excellent agreement between our DPs and the SCAN functional. Together, these findings confirm that the trained DPs accurately reproduce the energies and forces predicted by DFT (SCAN). Additional validation results, including detailed analysis, are presented in the SI.

■ ASSOCIATED CONTENT

Data Availability Statement

The following repository contains the three DPs used in this work, along with the training set, validation set, and input files required to train a DP potential. In addition, input files for the Quantum ESPRESSO, LAMMPS, and PLUMED program suites are included for reference. All data are available at: <https://github.com/sanghyunjonathan/Methanol-Adsorbed-TiO2-Interface>

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.5c07506>.

Parameters used in the simulation, additional details on the simulation methodology, and supporting data analysis (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

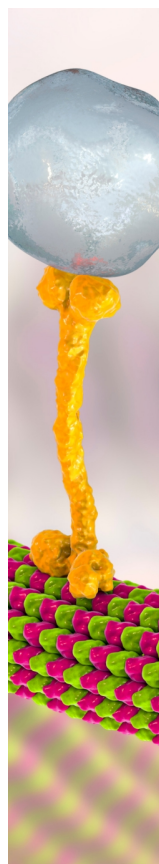
This work was supported by DOE BES, CSGB Division under Award DESC0007347. We used computing resources at the National Energy Research Scientific Computing Center (NERSC), under DOE Contract No. DE-AC02-05CH11231. We also acknowledge the use of Princeton Research Computing resources at Princeton University, which is a consortium of groups led by the Princeton Institute for Computational Science and Engineering (PICSciE) and the Office of Information Technology's Research Computing. The authors thank Dr. Axel Gomez for constructive and helpful discussions.

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